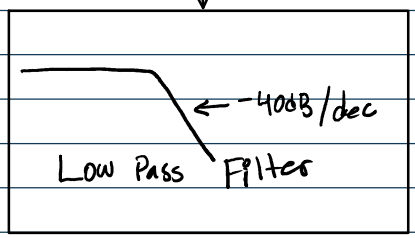
Power Supply Calculations

Sunday, October 25, 2026 4:55 PM

Power supply
- V_o = 30V max
- I_o = 5A max
- Noise 0.5%

LM317
- Good price
- Readily available
- V_o = 1.7 – 37V
- I_o = 1.5A
- 0.1% line & load reg
- eN = 0.003% 10Hz – 10kHz T = 25°C
- SVR = 80dB w/ 10NF Cadj 25°C, 120Hz

LM2596
- V_{in} = 60V max
- V_{out} = 1.2V – 37V
- low component count
- low cost
- 150KHz
- $V_o = V_{ref}\left(1 + \frac{R_2}{R_1}\right)$ $V_{ref} = 1.23$ $240\Omega < R_1 < 1.5K\Omega$



LM337
- I_o = 1.5A
- 77dB Ripple rejection (10NF)
- V_o = -1.25 – -37V
- RMS Noise = 0.003%
- $-V_o = -1.25\left(1 + \frac{R_2}{R_1}\right) + (-I_{adj} \times R_2)$

Recommended $R_1 = 120\Omega$

2 — 12V Channels (#1, #3, #4)

- For $V_{in} = 15V \rightarrow$ expect 150mV noise
-

$$SVR = 20 \cdot \log_{10}\left(\frac{V_{in}}{V_{out}}\right)$$

$$V_{out} = \frac{V_{in}}{10^{SVR/20}} = \frac{150mV}{10^{80/20}} = 15\mu V$$

$\boxed{15\,\mu V \text{ ripple}}$ (From bench power supply)

- LM317 Noise 0.003%. 10Hz – 10kHz

At $V_o = 12V$ max

$$eN = (12)(0.003\%) = 36\mu V$$

- $$\text{Ripple + Noise} = \sqrt{36\mu V^2 + 15\mu V^2} = 39\mu V_{rms}$$

- $39\mu V_{rms}$ can be high for analog signal processing w/ some exceptions

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- If op-amp is used w/ PSRR ~ 80

$$eN_{ps} = \frac{40\mu V}{10^{80/20}} = 4nV$$

- @ PSRR ~ 100

$$eN_{PS} = \frac{40\mu V}{10^{100/20}} = 0.4nV$$

-2 → -12V Output Channel (#2)

Power supply

@ 15V → 150mV noise

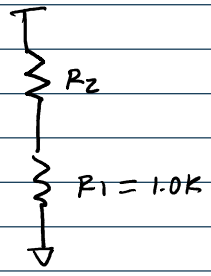
Switching Regulator

- $V_o = V_{ref}\left(1 + \frac{R_2}{R_1}\right)$ ← Set the output voltage

$$R_2 = \left(\frac{V_o}{V_{ref}} - 1\right) R_1$$

$$= \left(\frac{15}{1.23} - 1\right) \cdot 1K\Omega$$

$$\boxed{R_2 = 11.2K\Omega}$$



- Inductor selection
- From datasheet use 33N H, 3.5A
- Recommended C_1, R_1, R_2 start-up
  - C_m = 470nF, 50V
  - C_{out} = 220nF, 25V ← lower price & higher V rating
- Noise
- Capacitor Ripple

$$V_{cap} = \frac{I_o \cdot D}{C_{out} \cdot f} = \frac{(260mA)(0.50)}{(220nF)(150kHz)}$$

$V_{cap} = 3.9mV_{pp}$

• ESR Ripple

$V_{ESR} = I_{L_{peak}} \cdot ESR$

$\rightarrow Duty = \frac{15}{15+15.1} = \sim 0.50$

$\rightarrow I_{L_{avg}} = \frac{I_o}{1-D} = \frac{260mA}{1-0.5} = 0.52A$

• $I_{L,ripple} = \frac{V_{in} \cdot D}{L \cdot f} = \frac{15V \cdot 0.5}{33\mu H \cdot 150kHz} = 1.52A_{pp}$

• $I_{L,peak} = I_{L_{avg}} + \frac{I_{L,ripple}}{2}$

$= (0.52A) + \frac{1.52A_{pp}}{2}$

$= 1.28A$

$V_{ESR} = I_{L,peak} \cdot ESR$

$= (1.28A)(160m\Omega)$

$V_{ESR} = 0.2048V$

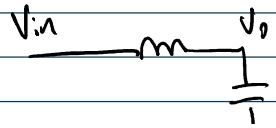
• Total noise from LM2596

$\Delta V_{tot} = V_{ESR} + V_{cap}$

$= 205mV + 3.9mV$

$\Delta V_{tot} = 208mV$

Low Pass Filte to filter out LM2596 regulator



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$$\frac{V_o}{V_{in}} = \frac{1}{1 - \omega^2 LC}$$

$$1 - \omega^2 LC = 0$$

$$\omega = \sqrt{\frac{1}{LC}}$$

$$f = \frac{1}{2\pi} \cdot \sqrt{\frac{1}{LC}}$$

for $L = 10nH$ & $C = 20nF$

$$f_c = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{10nH \cdot 20nF}} = 11.3kHz$$

@ f_c, we have $-3dB$

@ $-40dB/dec$ and $\frac{\frac{150kHz}{11.3kHz}}{10} = 1.33$ decade

Total attenuation $= -3 + (-40dB/dec) \cdot 1.33$

$= -56dB$

$$\text{Ripple out} = \frac{208mV}{10^{56/20}} = 330nV$$

- **LM337**
  - $V_{ripple} = \frac{V_{in}}{10^{PSRR/20}} = \frac{330nV}{10^{40/20}} = 3.3nV$
  - $eN = (12V)(0.003\%) = 360nV$
  - $eN_{tot} = \sqrt{360nV^2 + 3.3nV^2} = 360nV$

$\boxed{eN_{tot} = 360nV}$

@ Op-amp PSRR = 80

$$eN_{out} = \frac{360nV}{10^{80/20}} = 30.6\,pA$$

@ PSRR = 100

$$eN_{out} = \frac{360nV}{10^{100/20}} = 3.06\,pA$$